

Marilyn Braojos Gutierrez

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EDUCATION

Georgia Institute of Technology, Atlanta, GA, *GPA: 3.90/4.00*

Aug 2022 – Present

Ph.D. in Robotics – Aerospace Engineering

Expected Summer 2027

- **Selected Awards:** REACH Scholarship (2023 –), Goizueta Fellowship (2023 –), NASA Intl Travel Award (2024), Clare Boothe Luce (2026)

University of Florida, Gainesville, FL, *GPA: 3.91/4.00*

Aug 2015 – May 2022

M.S. in Mechanical Engineering (Dynamics, Systems, and Control)

May 2022

Bachelor of Science, Mechanical Engineering (Cum Laude Honors)

December 2020

Bachelor of Science, Psychology: Neuroscience

December 2020

- **Selected Awards:** McKnight Fellowship (2022), NASA FSGC MS Fellowship (2021–2022), Peterson Fellowship (2020)

INTERNSHIP EXPERIENCE

Aerospace Engineering Intern, NASA Jet Propulsion Lab, Pasadena, CA

May 2024 – Aug 2024

- Achieved <2 ns accuracy in clock drift corrections in GPS with transformer model architecture

Fault Protection Engineering Intern, NASA Jet Propulsion Lab, Pasadena, CA

May 2023 – May 2024

- Designed UI and functionality for the Mars Sample Return SRL mitigation matrix using FileMaker Pro system
- Developed FMECA and FTA for 12 subsystems in Mars Sample Return SRL to identify single point system failures

Robotics Intern, NASA Goddard Space Flight Center, Remote

June 2022 – Aug 2022

- Built Python application to control 6-DOF hexapod, mount, interferometer, and nanopositioners for LISA mission telescope tests
- Automated hexapod trajectory for LISA optical tests with Python API and QEMU emulator

Mechanical Engineering Intern, Raytheon Technologies, Springfield, VA

June 2021 – Nov 2021

- Designed 3 iterations of a pipe mating plate using SolidWorks to adapt a throttle body into a pipe system
- Completed trade study of 20+ OTS products for pipe system air regulation

Area Business Leader Co-Op, General Electric Appliances, Louisville, KY

Jan 2020 – May 2020

- Modeled common appliance parts with SolidWorks using GD&T and model-based definitions

RESEARCH EXPERIENCE

Graduate Research Assistant, Space Systems Design Lab, Atlanta, GA

Aug 2022 – Present

- Design and execute end-to-end machine learning pipeline for terrain identification robust to 3 sensor fault modes on rovers
- Developed deep learning architectures to achieve 6-hour time horizon satellite clock bias in GNSS-limited areas
- Led 31 Lunar Flashlight science operations contacts with the Deep Space Network

NASA Florida Space Grant Fellow, Precision Space Systems Lab, Gainesville, FL

Aug 2020 – May 2022

- Designed flight-ready fiber-coupled 2-lens UV LED system with SolidWorks and DataRay's WinCamD

TEACHING EXPERIENCE

Graduate Teaching Assistant, Georgia Institute of Technology, Atlanta, GA

Aug 2022 – July 2025

- **STEP Summer Camp (5/2025–7/2025):** Mentored 20+ groups of HS students prototyping a boat and rover using PVC and Arduino UNO
- **Spacecraft Engineering 1&2 (8/2024–5/2025):** Supervised 10 design project groups, of which 5 were demo-ed in microgravity conditions
- **Intro to AE (8/2022–12/2022):** Mentored 18 groups through the design and launch of a model rocket and glider

Undergraduate/Graduate Teaching Assistant, University of Florida, Gainesville, FL

Jan 2016 – May 2021

- **Mechanical Engineering Design 2&3 (8/2020–5/2021):** Head TA; Mentored 1 group directly through the design of a microbioreactor
- **Thermodynamics (5/2020–5/2021):** Lectured 70+ students on introductory thermodynamics concepts in 3 weekly discussion sessions
- **Control Systems Lab (8/2020–12/2020):** Lectured 45+ students on control schemes and debugged LabVIEW code
- **Pseudoscience (1/2017–12/2017):** Organized weekly discussion sessions for 30+ students about common pseudoscience practices
- **General Chemistry 1&2 (1/2016–5/2017):** Led 1-2 hour weekly discussions and exam reviews for 50+ students on basic chemistry topics

LEADERSHIP & INVOLVEMENT EXPERIENCE

- **CEED (8/2024–Present):** Head mentor; design workshops and directly advise 6 UG/3 grad AE students on graduate school preparation
- **Space Systems Design Lab (8/2023–5/2025):** Maintained lab website front-end
- **UF Shands Hospital (8/2016–5/2022):** Supervised 11 volunteers and directly assisted 5-6 patients in palliative care for 2 weekly shifts

PUBLICATIONS

[J3] Achieving Proprioceptive Terrain Classification with Deep Learning Models Robust to Sensor Faults on Lunar Rovers, *IEEE RAL*, **2026** [In Prep]

[J2] Predicting High Accuracy Long-term GPS Clock Bias with Autoregressive Deep Learning Architectures, *IEEE Access*, **2025** [Revision Submitted]

[C2] Developing Deep Learning Models to Predict Long-Term Satellite Clock Bias Corrections, *IAC*, **2024**

[J1] Lunar Flashlight Science Ground and Flight Measurements and Operations Using a Multi-Band Laser Reflectometer, *ICARUS*, **2024**

[C1] Short Wavelength UV LED Lens System to Attenuate Noise in LISA, *AIAA*, **2023**

SKILLS

- **Programming:** MATLAB, Python, LabVIEW, Minitab, C#
- **Software:** Amateur Radio Technician (Call Sign: KO4JOT), LaTeX, ROS2, PyTorch, Tensorflow, SolidWorks Certified Associate
- **Languages:** English (Native/Fluent), Spanish (Native/Fluent)